

How to Measure Legislative District Compactness If You Only Know it When You See it¹

Gary King²

Institute for Quantitative Social Science
Harvard University

University of Minnesota, 9/12/2019

¹Based on joint work with Aaron Kaufman and Mayya Komisarchik

²GaryKing.org

Redistricting Defines Democracy — & Needs Fixing

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation
 - \$100s of millions spent trying to influence the rules of the game

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation
 - \$100s of millions spent trying to influence the rules of the game
 - Litigation in almost every jurisdiction, every time

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation
 - \$100s of millions spent trying to influence the rules of the game
 - Litigation in almost every jurisdiction, every time
 - $\rightsquigarrow$ Get the ball, move the goalposts

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation
 - \$100s of millions spent trying to influence the rules of the game
 - Litigation in almost every jurisdiction, every time
 - $\rightsquigarrow$ Get the ball, move the goalposts
- Blamed for:

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy
 - Control redistricting $\rightsquigarrow$ Define basic units of representation
 - \$100s of millions spent trying to influence the rules of the game
 - Litigation in almost every jurisdiction, every time
 - $\rightsquigarrow$ Get the ball, move the goalposts
- Blamed for:
 - unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .
- How to fix this?

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via:

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via: Population equality,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via: Population equality, partisan fairness,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via: Population equality, partisan fairness, racial fairness,

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via: Population equality, partisan fairness, racial fairness, respect for municipal boundaries . . .

Redistricting Defines Democracy — & Needs Fixing

- Fundamental to Democracy

- Control redistricting $\rightsquigarrow$ Define basic units of representation
- \$100s of millions spent trying to influence the rules of the game
- Litigation in almost every jurisdiction, every time
- $\rightsquigarrow$ Get the ball, move the goalposts

- Blamed for:

- unfair elections, excessive partisanship, policy gridlock, partisan bias, lack of electoral responsiveness, racial bias, . . .

- How to fix this?

- Constrain redistricters via: Population equality, partisan fairness, racial fairness, respect for municipal boundaries . . . compactness

The Political Science Discipline & Redistricting

The Political Science Discipline & Redistricting

- Political science contributions to the real world

The Political Science Discipline & Redistricting

- Political science contributions to the real world
- Political science disconnect from the real world: Compactness

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
- Political science disconnect from the real world: Compactness

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - **Partisan fairness:** Invented standard (partisan symmetry) & methods
 - **Racial fairness:** Invented methods of ecological inference (for VRA)

- Political science disconnect from the real world: **Compactness**

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
- Political science disconnect from the real world: Compactness

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
- Political science disconnect from the real world: Compactness

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution:

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington:

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington: "Each district shall be as compact as possible"

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington: "Each district shall be as compact as possible"
 - Iowa:

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington: "Each district shall be as compact as possible"
 - Iowa: "avoid drawing districts that are oddly shaped"

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington: "Each district shall be as compact as possible"
 - Iowa: "avoid drawing districts that are oddly shaped"
 - Supreme Court:

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: “Legislative Districts shall be compact”
 - Washington: “Each district shall be as compact as possible”
 - Iowa: “avoid drawing districts that are oddly shaped”
 - Supreme Court: “One need not use Justice Stewart’s classic definition of obscenity—‘I know it when I see it’—... to recognize that *dramatically irregular shapes* may have sufficient probative force to call for an explanation”

The Political Science Discipline & Redistricting

- Political science contributions to the real world
 - Partisan fairness: Invented standard (partisan symmetry) & methods
 - Racial fairness: Invented methods of ecological inference (for VRA)
 - Forecasting elections in new districts, for all sides
 - Public service: as consultants, expert witnesses, special masters
 - Measurable impact: in numerous legal cases, state laws
- Political science disconnect from the real world: Compactness
 - Researchers: Assumed so complicated, numerous measures needed
 - Law: Assumed so simple, no definition needed!
 - Illinois Constitution: "Legislative Districts shall be compact"
 - Washington: "Each district shall be as compact as possible"
 - Iowa: "avoid drawing districts that are oddly shaped"
 - Supreme Court: "One need not use Justice Stewart's classic definition of obscenity—I know it when I see it"—... to recognize that *dramatically irregular shapes* may have sufficient probative force to call for an explanation"
 - Required in many other jurisdictions

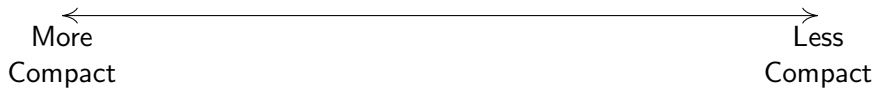
Compactness According to the Law

Compactness According to the Law

A simple single compactness dimension that you know when you see

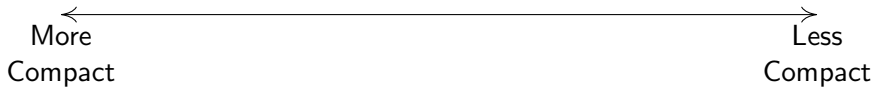
Compactness According to the Law

A simple single compactness dimension that you know when you see



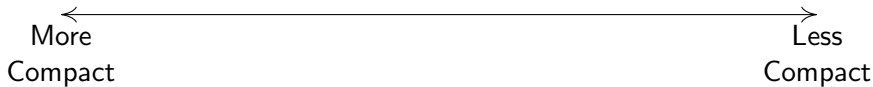
Compactness According to the Law

A simple single compactness dimension that you know when you see



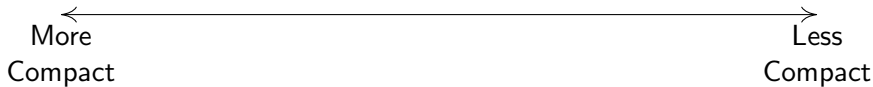
Compactness According to the Law

A simple single compactness dimension that you know when you see



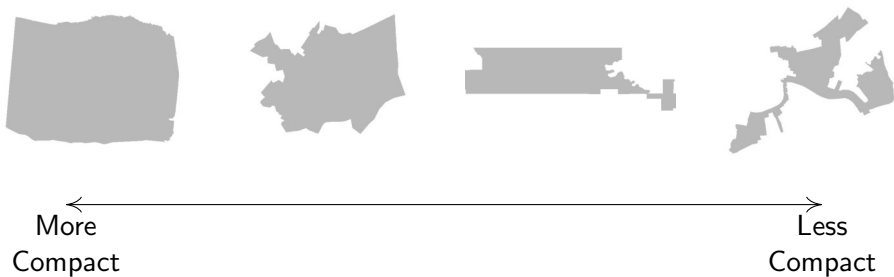
Compactness According to the Law

A simple single compactness dimension that you know when you see



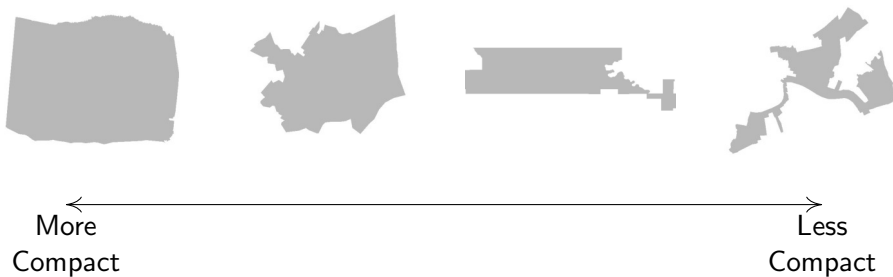
Compactness According to the Law

A simple single compactness dimension that you know when you see



Compactness According to the Law

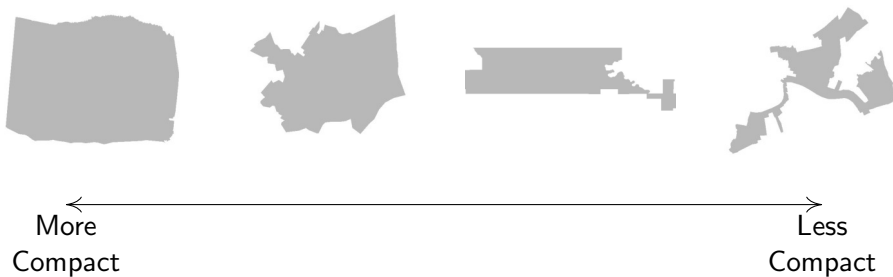
A simple single compactness dimension that you know when you see



- The dimension **is** intuitive

Compactness According to the Law

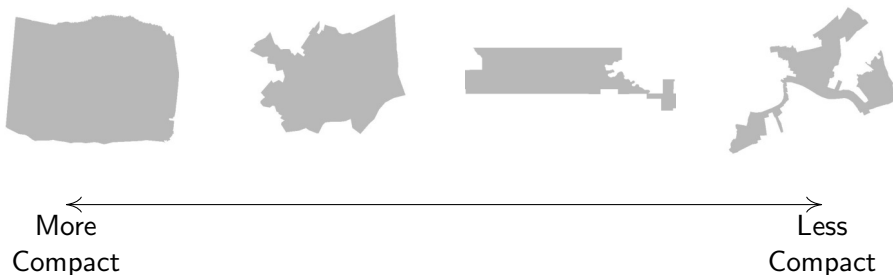
A simple single compactness dimension that you know when you see



- The dimension **is** intuitive
- How to **estimate** where a new district shape falls on this dimension?

Compactness According to the Law

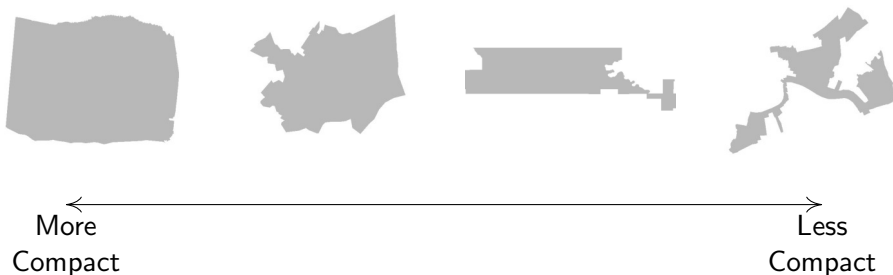
A simple single compactness dimension that you know when you see



- The dimension **is** intuitive
- How to **estimate** where a new district shape falls on this dimension?
- Only a **consensus measure** can constrain advocates

Compactness According to the Law

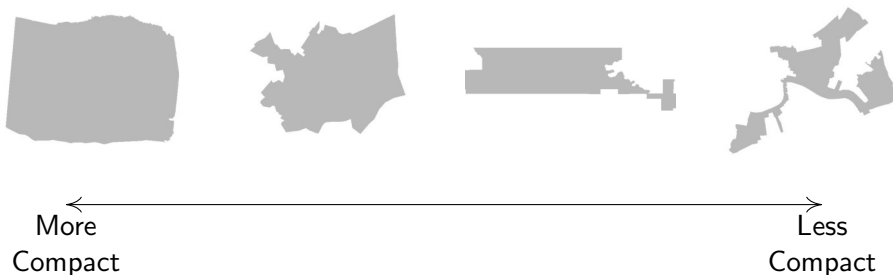
A simple single compactness dimension that you know when you see



- The dimension **is** intuitive
- How to **estimate** where a new district shape falls on this dimension?
- Only a **consensus measure** can constrain advocates
- Dimension **relative** to geography;

Compactness According to the Law

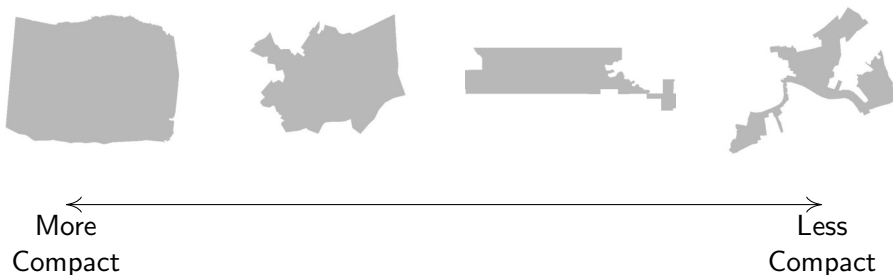
A simple single compactness dimension that you know when you see



- The dimension **is** intuitive
- How to **estimate** where a new district shape falls on this dimension?
- Only a **consensus measure** can constrain advocates
- Dimension **relative** to geography; could **generalize** (e.g., population)

Compactness According to the Law

A simple single compactness dimension that you know when you see



- The dimension **is** intuitive
- How to **estimate** where a new district shape falls on this dimension?
- Only a **consensus measure** can constrain advocates
- Dimension **relative** to geography; could **generalize** (e.g., population)
- **↪ Let's start with existing measures by social scientists**

Measure 1: Length/Width Ratio of Min Bounding Box

Measure 1: Length/Width Ratio of Min Bounding Box

Squarish districts more compact than long thin ones

Measure 1: Length/Width Ratio of Min Bounding Box

Squarish districts more compact than long thin ones



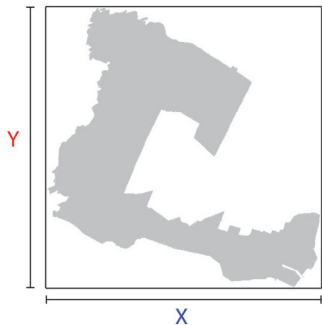
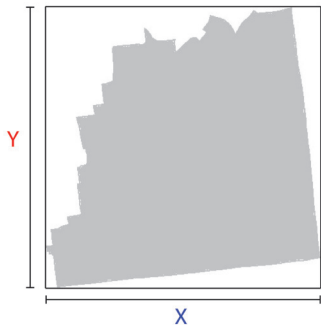
Measure 1: Length/Width Ratio of Min Bounding Box

Squarish districts more compact than long thin ones



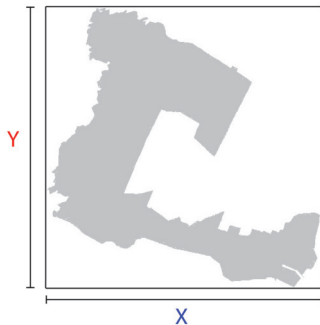
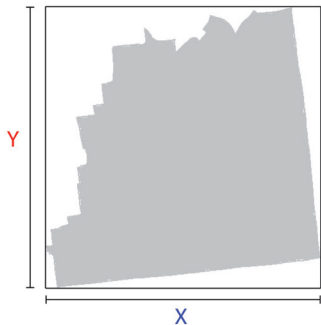
Measure 1: Length/Width Ratio of Min Bounding Box

Squarish districts more compact than long thin ones



Measure 1: Length/Width Ratio of Min Bounding Box

Squarish districts more compact than long thin ones



In both districts: $X/Y \approx 1.30$

Measure 2: Reock, District / Bounding Circle Areas

Measure 2: Reock, District / Bounding Circle Areas

Circular districts are most compact

Measure 2: Reock, District / Bounding Circle Areas

Circular districts are most compact



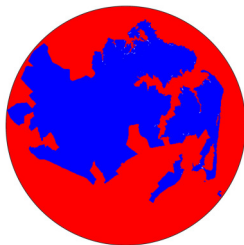
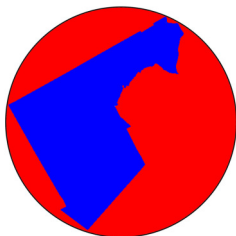
Measure 2: Reock, District / Bounding Circle Areas

Circular districts are most compact



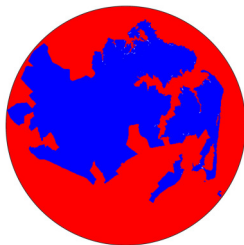
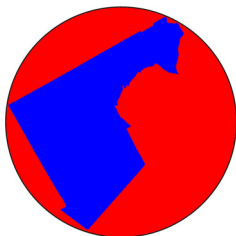
Measure 2: Reock, District / Bounding Circle Areas

Circular districts are most compact



Measure 2: Reock, District / Bounding Circle Areas

Circular districts are most compact



In both cases, $X/(Y + X) \approx 0.37$

Measure 3: Boyce-Clark, Variation in Centroid Deviations

Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar

Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar



Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar



Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar



Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar



Measure 3: Boyce-Clark, Variation in Centroid Deviations

All travel distances from center should be similar



In both cases, $\text{MAD}(r)/\bar{r} \approx 0.31$

A Brief Rotational Invariance Interlude:

A Brief Rotational Invariance Interlude: Can you Name this Celebrity?

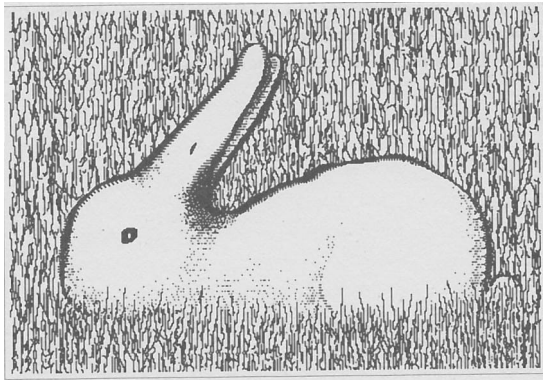
A Brief Rotational Invariance Interlude: Can you Name this Celebrity?



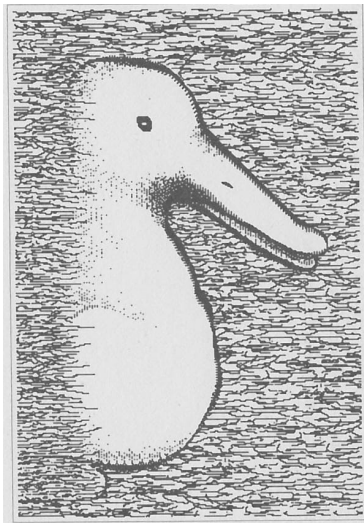
A Brief Rotational Invariance Interlude: Can you Name this Celebrity?



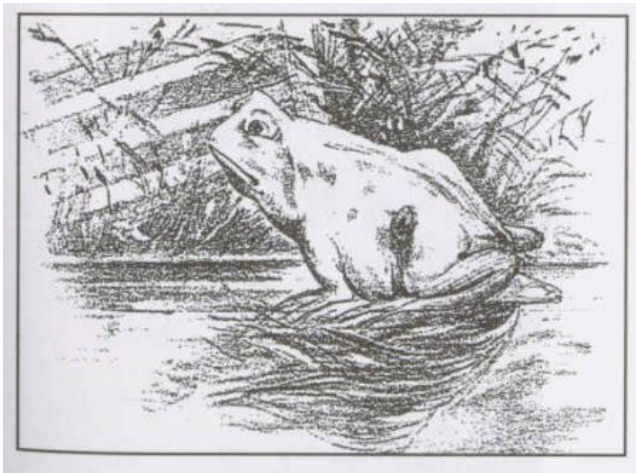
A Brief Interlude on Perception: See the Rabbit?



A Brief Interlude on Perception: See the Rabbit Duck?



A Brief Interlude on Perception: See the Frog?



A Brief Interlude on Perception: See the Frog Horse?



Human Perception: Not Rotationally Invariant

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed
 - Almost all are rotationally invariant

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed
 - Almost all are rotationally invariant
 - Blind to what humans perceive

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed
 - Almost all are rotationally invariant
 - Blind to what humans perceive
- Which is more compact?

Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed
 - Almost all are rotationally invariant
 - Blind to what humans perceive
- Which is more compact?



Human Perception: Not Rotationally Invariant

- Existing measures of compactness:
 - Nearly 100 proposed
 - Almost all are rotationally invariant
 - Blind to what humans perceive
- Which is more compact?



- ↪ Measuring “you know it when you see it”: No rotational invariance

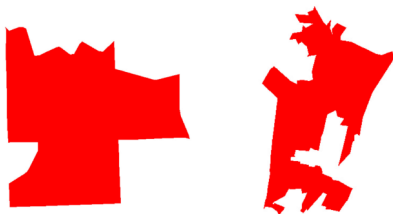
New Measure: Y-Symmetry, area of symmetric reflection

New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact

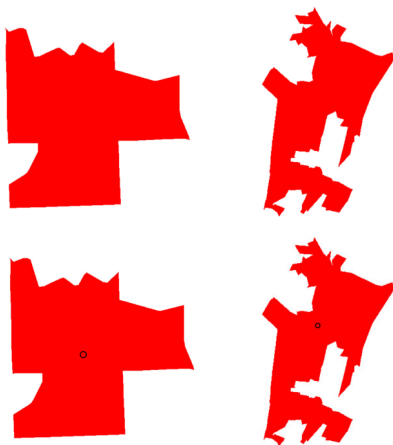
New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact



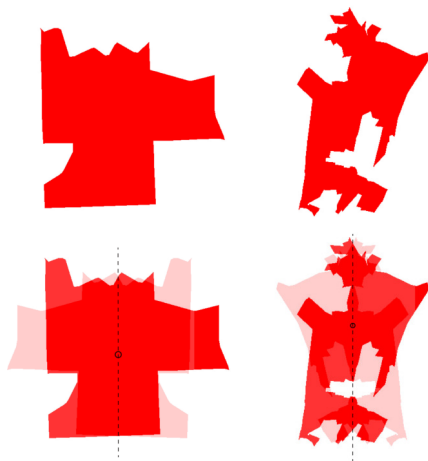
New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact



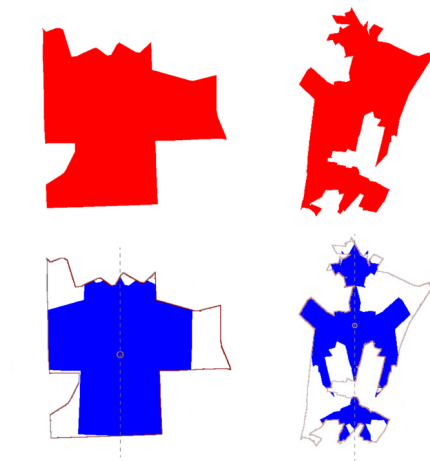
New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact



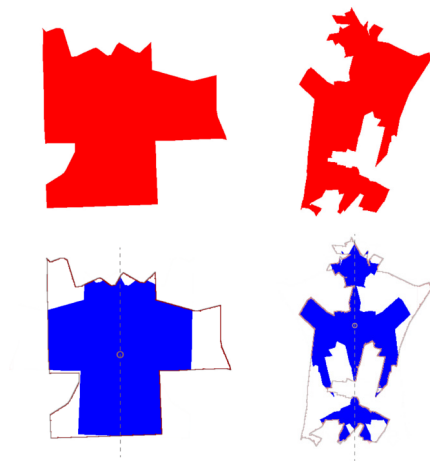
New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact



New Measure: Y-Symmetry, area of symmetric reflection

Symmetric figures (circles, squares) are more compact



In both cases, $\text{Overlap} / \text{Original Area} \approx 0.34$

New Measure 2: Number of Visually Significant Corners

New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

~> Fewer corners is more compact

New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

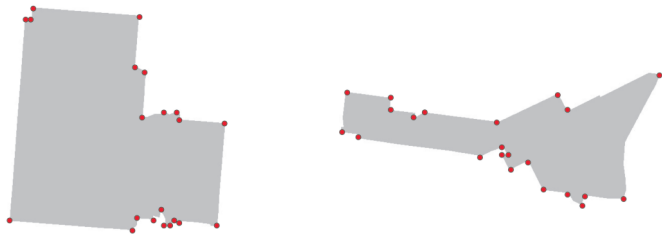
~> Fewer corners is more compact



New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

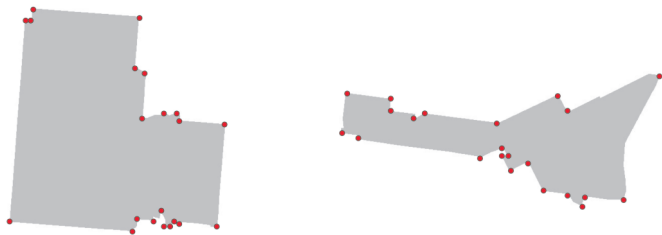
~> Fewer corners is more compact



New Measure 2: Number of Visually Significant Corners

Computer vision algorithm identifies “objects” in photos

~> Fewer corners is more compact



Both districts have 21 significant corners

Which is more compact?



Which is more compact? Depends on the standard!



Which is more compact? Depends on the standard!



Convex Hull

4

3

2

1

Which is more compact? Depends on the standard!



Convex Hull

4

3

2

1

Reock

1

2

3

4

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures;

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures; 7 unique rankings

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures; 7 unique rankings
- Unusual?

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures; 7 unique rankings
- **Unusual?** From 18,215 Congressional and State Legislative Districts,

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures; 7 unique rankings
- **Unusual?** From 18,215 Congressional and State Legislative Districts, we found 162 trillion others (about 0.15%)

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
Reock	1	2	3	4
Polsby-Popper	4	1	2	3
Boyce-Clark	2	3	1	4
Length/Width	3	2	1	4
X-Axis Symmetry	1	4	3	2
Significant Corners	4	1	3	2

- 7 measures; 7 unique rankings
- **Unusual?** From 18,215 Congressional and State Legislative Districts, we found 162 trillion others (about 0.15%)
- **Many more inconsistencies on individual districts**

Spanning the Academic–Legal Divide

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:
 - Know it when they see it,

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:
 - Know it when they see it,
 - See the same dimension

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:
 - Know it when they see it,
 - See the same dimension
 - I.e., estimate the one dimension of legal interest; show it has:

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:
 - Know it when they see it,
 - See the same dimension
 - I.e., estimate the one dimension of legal interest; show it has:
 - high intercoder (and intracoder) reliability

Spanning the Academic–Legal Divide

- (Recall) The concept of compactness
 - Researchers: So complicated, numerous measures needed
 - Law: So simple, no definition needed
- Our Hypothesis: both are right
 - The Theoretical Concept: multidimensional
 - The Legal Concept: one dimensional
 - Which dimension? The one we know when we see
- How do we know if we find it?
 - Public officials and many other types of people:
 - Know it when they see it,
 - See the same dimension
 - I.e., estimate the one dimension of legal interest; show it has:
 - high intercoder (and intracoder) reliability
 - high predictive accuracy

How to rank districts on the same dimension?

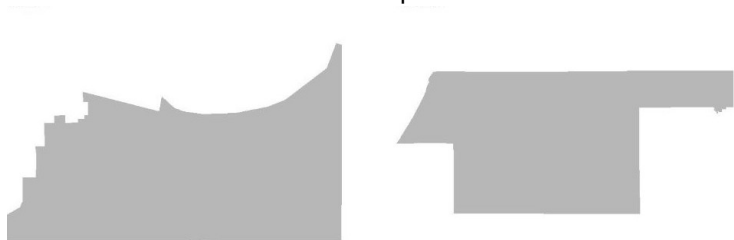
How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

Paired Comparison



How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

Paired Comparison



Utterly fails on inter- and intra-coder reliability

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

Full Ranking

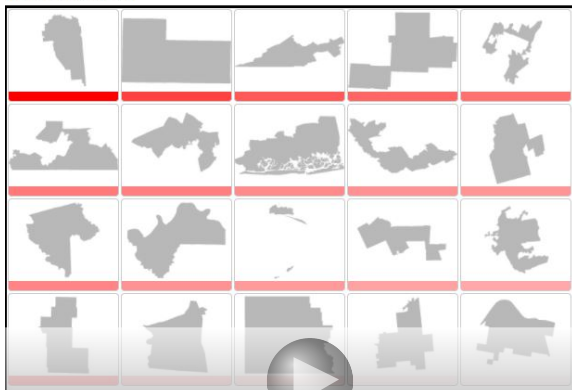


How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

Full Ranking — on line

MOST Compact Here



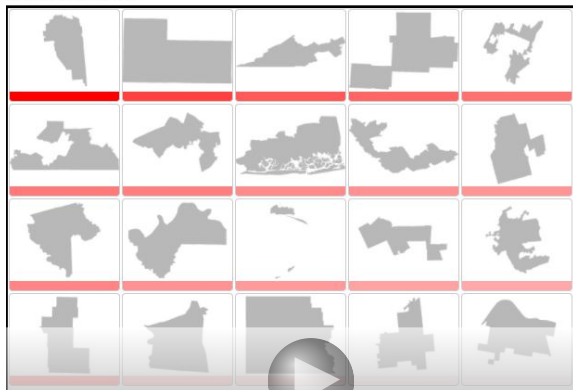
LEAST Compact Here

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

Full Ranking — on line

MOST Compact Here



LEAST Compact Here

We show: very high reliability

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?
 - 190 paired comparisons is tedious and boring;

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?
 - 190 paired comparisons is tedious and boring;
Ranking is more intellectually engaging

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?
 - 190 paired comparisons is tedious and boring;
Ranking is more intellectually engaging
 - Saves time: 1 task v 190 comparisons

How to rank districts on the same dimension?

Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?
 - 190 paired comparisons is tedious and boring;
Ranking is more intellectually engaging
 - Saves time: 1 task v 190 comparisons
 - Paired Comparisons can be answered on **different dimensions**

How to rank districts on the same dimension?

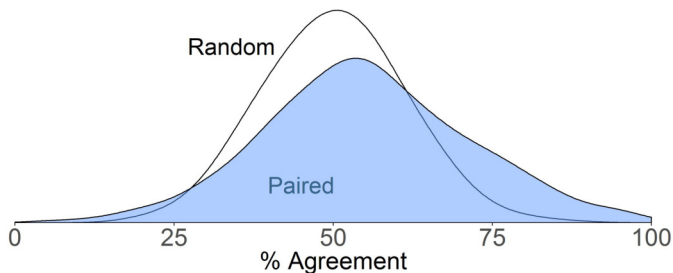
Paired Comparisons (Fechner 1860; Thurstone 1912) v Ranking (very old, rarely used)

- Why Paired Comparisons is supposedly better
 - Everyone does what they are good at:
 - Respondents answer simple, concrete questions
 - Researchers reconstruct the scale
 - Much easier: $\binom{20}{2} = 190$ pairs v $20! \approx 2$ quintillion ranks
- Why Ranking is actually better (at least in our application)
 - Humans use time-saving heuristics.
Would it take you 2 quintillion seconds to rank 20 districts?
 - 190 paired comparisons is tedious and boring;
Ranking is more intellectually engaging
 - Saves time: 1 task v 190 comparisons
 - Paired Comparisons can be answered on **different dimensions**
Ranking: all evaluations on **one dimension** of user's choice

Intercoder Reliability of Pairs

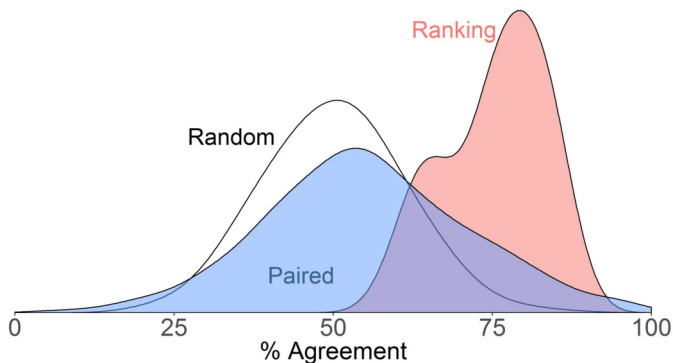
Intercoder Reliability of Pairs

Paired Comparisons: only slightly better than chance;



Intercoder Reliability of Pairs

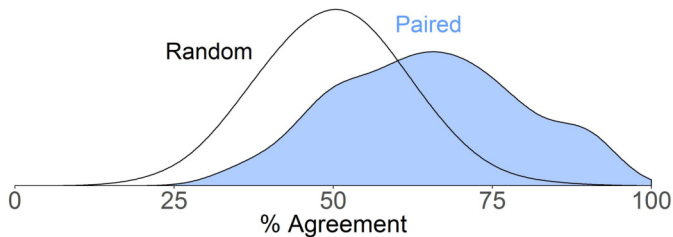
Paired Comparisons: only slightly better than chance; Pairs implied by ranks: better



Intracoder Reliability of Pairs

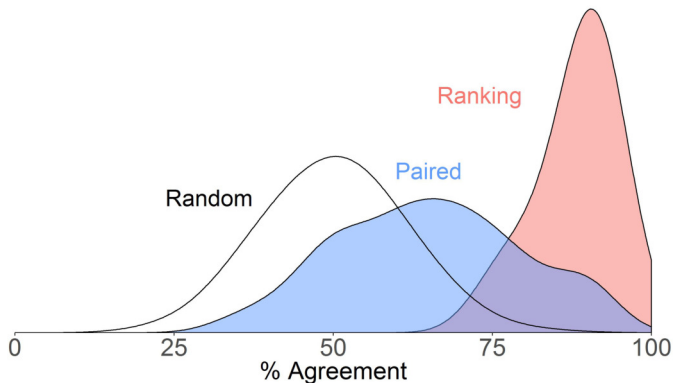
Intracoder Reliability of Pairs

Paired Comparisons: better than chance;

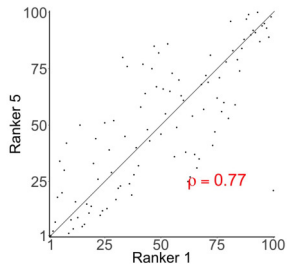


Intracoder Reliability of Pairs

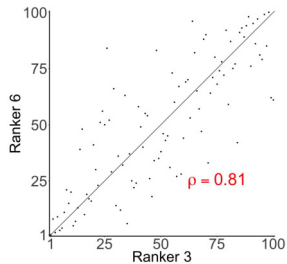
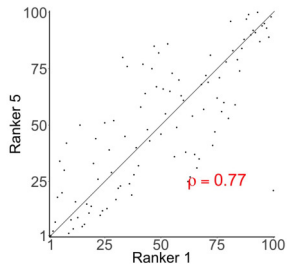
Paired Comparisons: better than chance; Pairs implied by ranks: much better



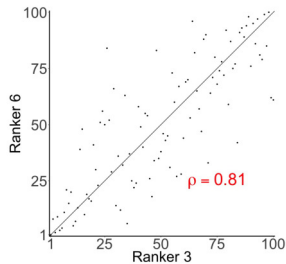
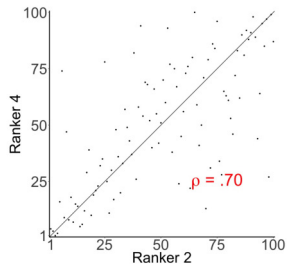
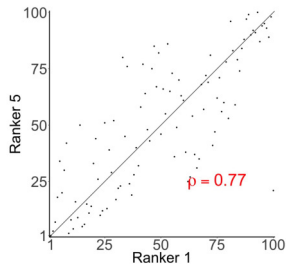
Intercoder Reliability on Ranks



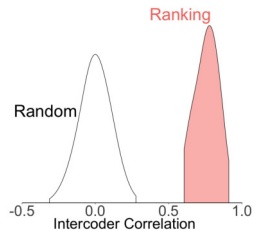
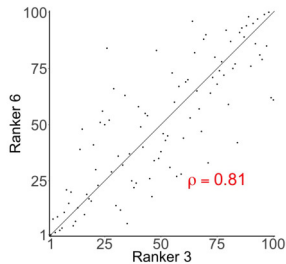
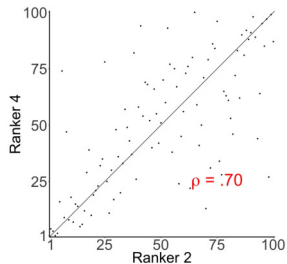
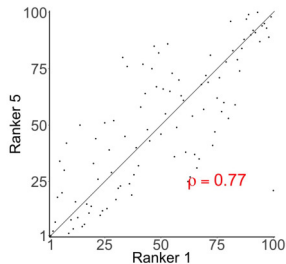
Intercoder Reliability on Ranks



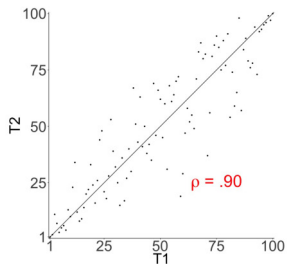
Intercoder Reliability on Ranks



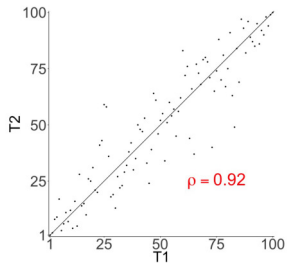
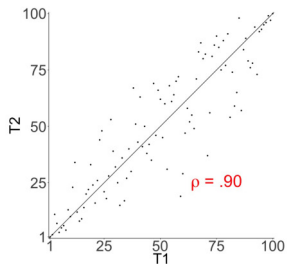
Intercoder Reliability on Ranks



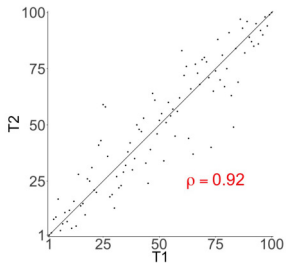
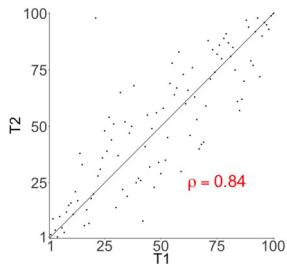
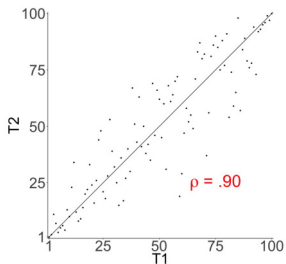
Intracoder Reliability on Ranks



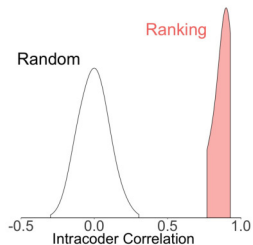
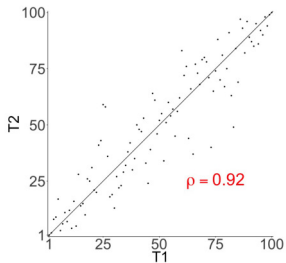
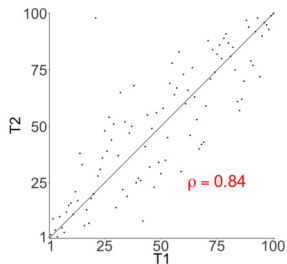
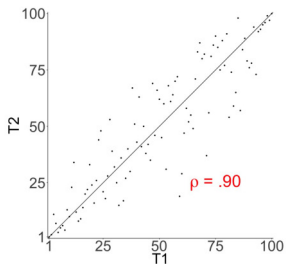
Intracoder Reliability on Ranks



Intracoder Reliability on Ranks



Intracoder Reliability on Ranks



So we can measure it. Can we model it?

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- Training data: Outcome variable from human rankings

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- Training data: Outcome variable from human rankings
- Outcome measure: A district's rank (in a set of 100)

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- Training data: Outcome variable from human rankings
- Outcome measure: A district's rank (in a set of 100)
- Covariates. Features of district shape

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- Training data: Outcome variable from human rankings
- Outcome measure: A district's rank (in a set of 100)
- Covariates. Features of district shape
 - Existing: Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**
 - **Polanyi's Paradox:**

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**
 - **Polanyi's Paradox:** we know more than we can tell

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**
 - **Polanyi's Paradox:** we know more than we can tell
 - **Tell!**

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**
 - **Polanyi's Paradox:** we know more than we can tell
 - **Tell!** squarish, with minimal arms, pockets, islands, or jagged edges

So we can measure it. Can we model it?

Goal: Compactness score = $f(\text{shape})$

- **Training data:** Outcome variable from human rankings
- **Outcome measure:** A district's rank (in a set of 100)
- **Covariates. Features of district shape**
 - **Existing:** Reock, Polsby-Popper, Convex Hull, Length/Width, Boyce-Clark...
 - **Geometric:** Perimeter, area, vertices, polygons, vertex variance, edge length variance...
 - **New:** X-axis symmetry, Y-axis symmetry, Significant Corners...
- **Ensemble of predictive methods:** least squares, AdaBoosted decision trees, SVM, random forests...
- **Meaning of resulting measure:**
 - **Polanyi's Paradox:** we know more than we can tell
 - **Tell!** squarish, with minimal arms, pockets, islands, or jagged edges
 - (Not a description of any one existing measure)

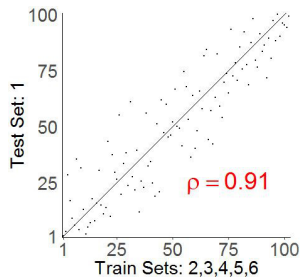
Model Validation: 6-Fold Cross-validation

Model Validation: 6-Fold Cross-validation

Predict Test Set from 5 Training Sets

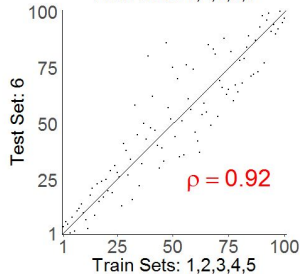
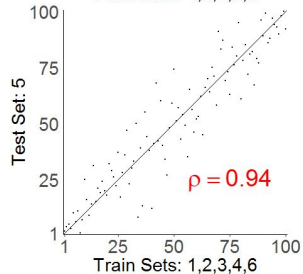
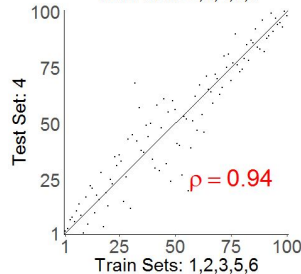
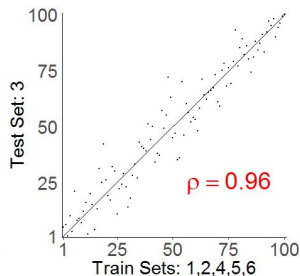
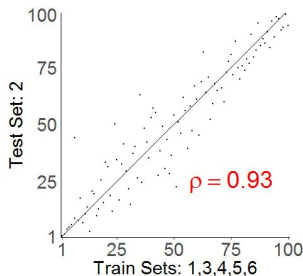
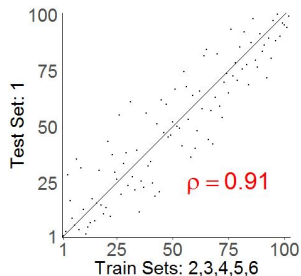
Model Validation: 6-Fold Cross-validation

Predict Test Set from 5 Training Sets



Model Validation: 6-Fold Cross-validation

Predict Test Set from 5 Training Sets



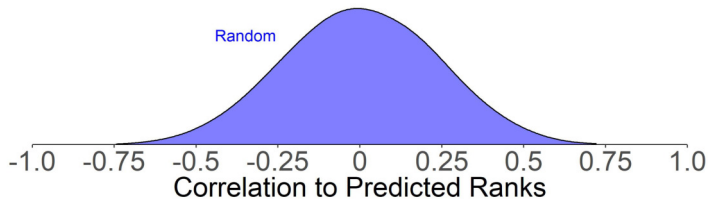
Model Validation: Diverse Respondents

Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting

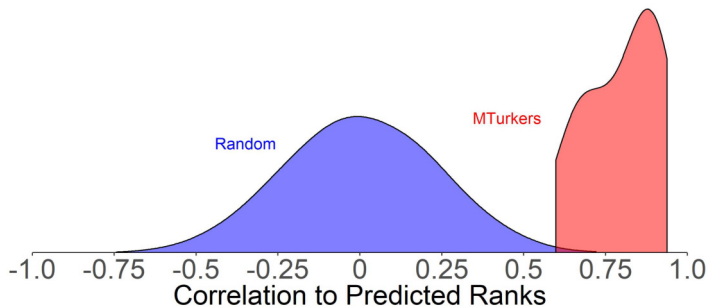
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



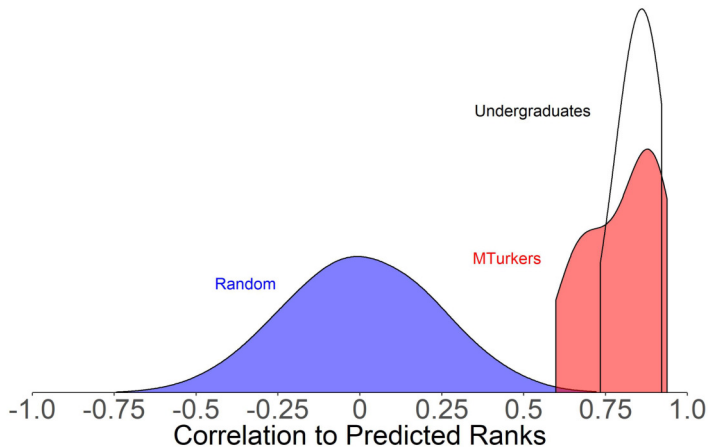
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



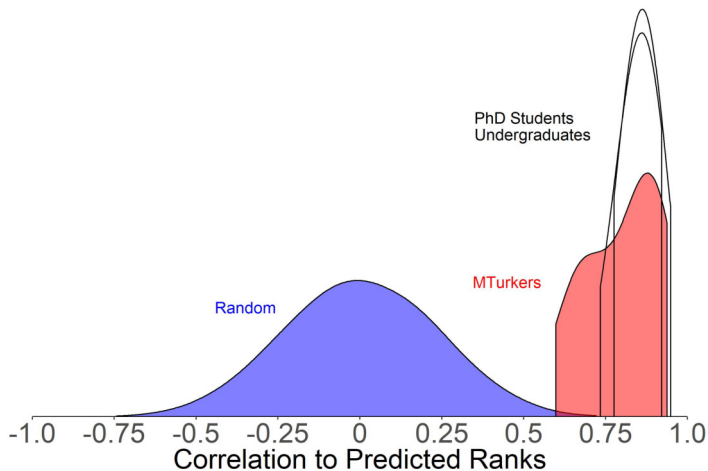
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



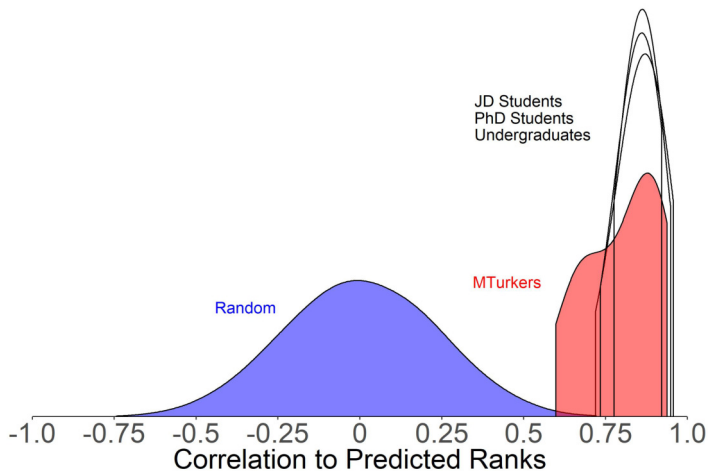
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



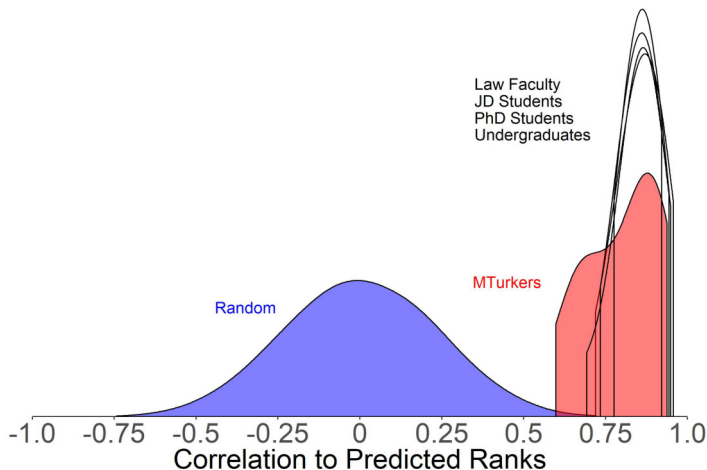
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



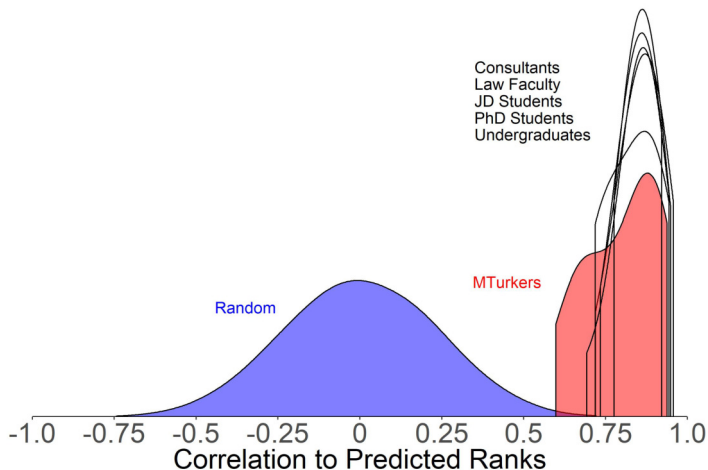
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



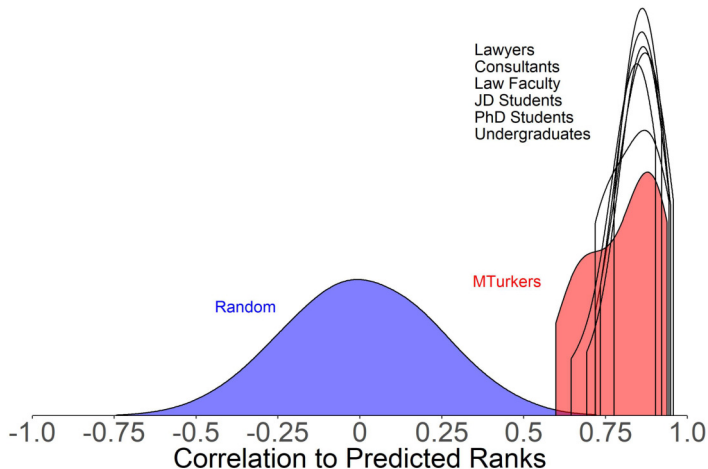
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



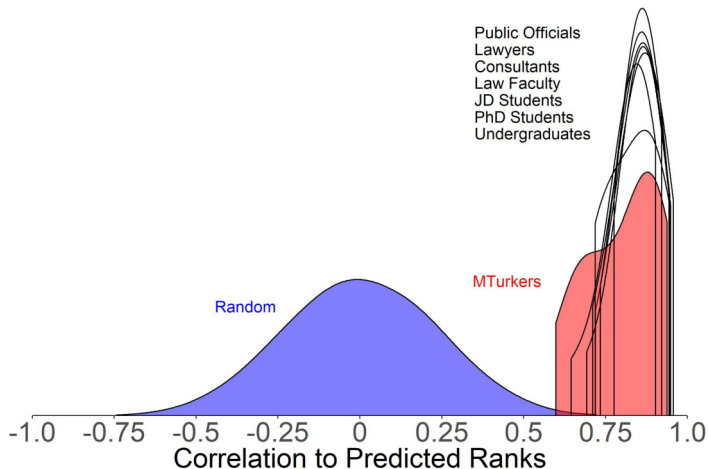
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



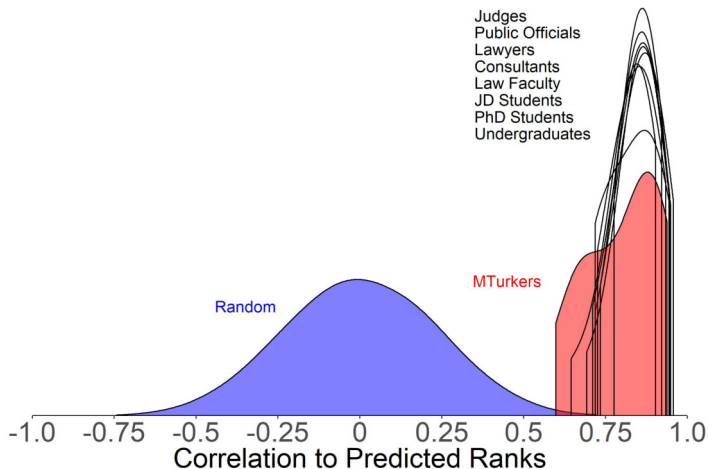
Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



Model Validation: Diverse Respondents

Respondents ranging from ordinary citizens to those responsible for redistricting



What do you think?

What do you think?

Our measure:	COMPACT	noncompact	noncompact	COMPACT
Existing measure:	COMPACT	noncompact	COMPACT	noncompact

What do you think?

Our measure:

COMPACT

noncompact

noncompact

COMPACT

Existing measure:

COMPACT

noncompact

COMPACT

noncompact

Reock



What do you think?

Our measure:	COMPACT	noncompact	noncompact	COMPACT
Existing measure:	COMPACT	noncompact	COMPACT	noncompact

Reock



Boyce-Clark



What do you think?

	Our measure: Existing measure:	COMPACT COMPACT	noncompact noncompact	noncompact COMPACT	COMPACT noncompact
Reock					
Boyce-Clark					
Length/Width					

What do you think?

	Our measure: Existing measure:	COMPACT COMPACT	noncompact noncompact	noncompact COMPACT	COMPACT noncompact
Reock					
Boyce-Clark					
Length/Width					
X-Symmetry					

What do you think?

Our measure:	COMPACT	noncompact	noncompact	COMPACT
Existing measure:	COMPACT	noncompact	COMPACT	noncompact

Reock



Boyce-Clark



Length/Width



X-Symmetry



Concluding Remarks

Concluding Remarks

- We address: Disconnect between political science & the real world

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**
 - Measures: for 18,215 Congressional & State Legislative districts

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**
 - Measures: for 18,215 Congressional & State Legislative districts
 - Software to calculate compactness from any district shape

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**
 - Measures: for 18,215 Congressional & State Legislative districts
 - Software to calculate compactness from any district shape
- **Along the way:**

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**
 - Measures: for 18,215 Congressional & State Legislative districts
 - Software to calculate compactness from any district shape
- **Along the way:**
 - New perspective on > 150 year consensus of ranking v paired comparisons

Concluding Remarks

- **We address:** Disconnect between political science & the real world
 - The Theoretical Concept: multidimensional and complex
 - The Legal Concept: one dimensional and simple
- **A proposed resolution:** measure the one dimension everyone sees
 - Calculated solely from district geometry
 - Very high intercoder & intracoder reliability
 - Very high predictive validity
 - Diverse people see it the same way
 - $\rightsquigarrow$ Continue political science tradition of contributing to a fundamental part of representative democracy
- **Accompanying this paper:**
 - Measures: for 18,215 Congressional & State Legislative districts
 - Software to calculate compactness from any district shape
- **Along the way:**
 - New perspective on > 150 year consensus of ranking v paired comparisons
 - New directions for two venerable literatures

For more information



AaronRKaufman.com



GaryKing.org



MayyaKomisarchik.com

Paper, data, software, slides: j.mp/Compactness